Prediction of Progression-Free Survival in Breast Cancer Patients

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2026-03-23

Introduction

Progression-free survival (PFS) is a key clinical endpoint in breast cancer research, defined as the time from diagnosis to disease progression or death. In this report, we build a prediction model for log-transformed PFS using data from 2,000 breast cancer patients diagnosed between 1990 and 1995 (Part I), and evaluate whether this model generalizes to a separate cohort of 2,000 patients diagnosed between 2000 and 2005 (Part II).

We consider four candidate models that span a range of complexity: **elastic net** as a regularized linear baseline with built-in variable selection; **PLS** as a dimension-reduction approach suited to correlated predictors; **MARS** for automatically capturing nonlinearity and interactions while maintaining interpretable piecewise-linear structure; and **GAM** for flexible nonparametric smoothing of continuous predictors. This set covers the spectrum from linear to nonlinear methods, allowing us to assess whether additional model flexibility translates into meaningful predictive gains.

Part I: Prediction Model for PFS (1990–1995 Cohort)

Data Overview

The dataset (datA) contains 2,000 patients with complete follow-up and no missing values. The response variable is log-transformed PFS (months). There are 15 predictors: five continuous (age,

BMI, tumor size, number of positive lymph nodes, Ki-67 index) and ten categorical (menopausal status, race, tumor grade, ER/PR/HER2 status, clinical stage, surgery type, chemotherapy, and hormonal therapy). Table 1 summarizes the sample characteristics. Exploratory analysis revealed that log(PFS) is approximately normally distributed (Figure 1). Among continuous predictors, tumor size and lymph node count show negative associations with log(PFS), while correlations among predictors are generally low (Figures 2–3). Boxplots indicate that ER-positive status, lower tumor grade, and earlier clinical stage are associated with longer PFS (Figure 4).

Model Training

All analyses were conducted in R (version 4.5) using the `caret` package as the primary modeling framework, with `glmnet` for elastic net, `earth` for MARS, `mgcv` for GAM, and `pls` for PLS. Categorical predictors (menopause, race, tumor grade, ER, PR, HER2, stage, surgery, chemo, hormonal therapy) were converted to factors, and the ten resulting factor variables were expanded into dummy indicators via treatment coding (first level as reference). All models were evaluated using 10-fold CV with the same pre-specified fold indices (generated by `createFolds` with seed 8106) to ensure a fair comparison. The tuning parameter selection criterion was minimization of CV RMSE. After identifying the best hyperparameters, each final model was refit on the entire training set (datA, $n = 2000$).

Elastic net was tuned over a grid of mixing parameter $\alpha \in \{0, 0.1, \dots, 1\}$ (11 values from ridge to lasso) and regularization parameter λ over 100 values equally spaced on the log scale from e^{-6} to e^{2}. No additional centering or scaling was applied, as `glmnet` standardizes internally. The optimal model selected $\alpha = 1$ (pure Lasso) with $\lambda = 0.0135$, retaining 11 non-zero coefficients.

PLS was tuned over 1–15 components. Predictors were centered and scaled within each CV fold via `preProcess`. The best model used 10 components.

MARS was tuned over interaction degree $\in \{1, 2, 3\}$ and the maximum number of retained terms (nprune) $\in \{2, \dots, 18\}$, with internal generalized cross-validation (GCV) used for basis function pruning. The best model selected degree 1 (additive, no interactions) with nprune = 10, yielding 9 basis functions involving 7 predictors.

GAM was fitted via `mgcv::gam()` with thin-plate regression splines (`s()`) on the five continuous predictors (age, BMI, tumor size, lymph nodes, Ki-67) and parametric (linear) terms for all categorical predictors. Smoothing parameters were estimated by restricted maximum likelihood (REML). Since `caret`'s GAM wrapper applies smoothing to all columns including dummy variables, we instead performed manual 10-fold CV using the same fold indices to maintain proper handling of categorical terms.

Results

Cross-validated performance was similar across all four models (Table 2, Figure 5). GAM achieved the lowest CV RMSE (0.699), followed closely by elastic net (0.701), PLS (0.703), and MARS (0.703). The differences in RMSE are less than 1%, suggesting that additional model flexibility provides negligible predictive gain. The modest R^2 (approximately 0.20) indicates that the available predictors explain a limited proportion of PFS variability, which is common in clinical prediction settings with heterogeneous patient populations.

Final model selection. Given that all four models perform nearly identically, we select MARS as the final model based on the principle of parsimony and interpretability. MARS uses only 9 piecewise-linear terms involving 7 predictors, producing a fully transparent model that can be written as an explicit equation and communicated to clinicians. By contrast, GAM's smooth functions, while slightly more accurate, cannot be easily summarized in closed form. The Lasso retains 11 coefficients but assumes strictly linear effects, missing the nonlinear patterns MARS captures (e.g., the BMI knot at 23.9). PLS, with 10 latent components, offers neither the interpretability of MARS nor a clear predictive advantage.

Model interpretation. The Lasso coefficients (Table 3) identify the most influential predictors: ER status (+0.225), HER2 status (–0.144), hormonal therapy (+0.120), tumor grade 3 (–0.103), lymph nodes (–0.060), and tumor size (–0.048). The MARS model (Table 4) confirms these findings with additional nonlinear detail: log(PFS) increases by 0.079 for each unit decrease in lymph nodes below 6, decreases by 0.049 per cm of tumor size, and shows a BMI effect with a knot at 23.9 kg/m^2. ER-positive status increases log(PFS) by 0.260, while HER2-positive and poorly differentiated tumors decrease it by 0.186 and 0.170, respectively.

Variable importance rankings from both models consistently place lymph nodes, ER status, tumor grade, HER2 status, and tumor size as the top predictors (Figure 6). Partial dependence plots (Figure 7) and GAM smooth functions (Figure 8) further illustrate these relationships. Residual diagnostics (Figure 9) show no major departures from model assumptions.

In summary, we recommend the MARS model for predicting PFS in this cohort. The key predictors are the number of positive lymph nodes, ER/HER2 receptor status, tumor grade, tumor size, and hormonal therapy receipt—consistent with established breast cancer prognostic factors.

Part II: Generalization to the 2000–2005 Cohort

Evaluating Model A on Data B

We applied all four models trained on datA to datB. As shown in Table 5, RMSE increased from approximately 0.70 (CV on datA) to approximately 0.84 on datB, representing a relative increase of roughly 20%. The R^2 dropped from approximately 0.20 to approximately 0.15. The observed-versus-predicted plot (Figure 10) reveals systematic underprediction for patients with longer PFS, and the residual distribution (Figure 11) is shifted positively (mean residual > 0), indicating that patients in the later cohort tend to have longer PFS than predicted by the 1990s model. These results demonstrate that the prediction model does not generalize well to the 2000–2005 cohort.

Understanding the Distributional Shift

Comparing the two cohorts (Table 6, Figures 12–13), log(PFS) in cohort B has a higher mean, reflecting improvements in breast cancer treatment over the decade. Treatment patterns also shifted—notably increased hormonal therapy use—and the distributions of clinical stage and other variables changed. These temporal shifts explain the poor transportability.

New Model for the 2000–2005 Cohort

We retrained all four models on datB using the same tuning grids and a new set of pre-specified 10-fold CV indices. The best performer was again GAM (CV RMSE = 0.669, R^2 = 0.181), followed closely by MARS (CV RMSE = 0.671, R^2 = 0.177; see Table 7 and Figure 14). Following the same parsimony rationale as Part I, we again recommend MARS as the final model for cohort B, as its CV RMSE is within 0.5% of GAM while providing an explicit, interpretable equation. Key differences compared to Model A:

1. **Interaction effects**: MARS for cohort B selected degree 2 (interactions), identifying interactions between tumor size and lymph nodes, tumor size and Ki-67, and BMI with both ER status and hormonal therapy—suggesting a more complex prognostic landscape.

2. **Hormonal therapy**: Its MARS coefficient increased from +0.137 to +0.425, and from +0.120 to +0.173 in elastic net (Table 8), likely reflecting increasing efficacy of targeted hormonal therapies (e.g., aromatase inhibitors) in the 2000s.

3. **HER2 status**: HER2 was a significant negative predictor in cohort A (Lasso: –0.144; MARS: –0.186) but attenuated in cohort B (Lasso: +0.029; absent from MARS), aligning with the introduction of trastuzumab (Herceptin) around 2000.

4. **Variable importance shift**: Lymph nodes was the top predictor in cohort A; in cohort B, tumor size and Ki-67 emerged as the top two, with hormonal therapy third (Figure 15).

5. **Broader predictor set**: Elastic net for cohort B retained 15 non-zero coefficients (vs. 11), newly including menopause, PR status, and clinical stage.

In conclusion, the 1990–1995 model does not generalize well to the 2000–2005 cohort. The changes—enhanced role of hormonal therapy, diminished HER2 impact, and emergence of interaction effects—reflect genuine shifts in breast cancer treatment. A cohort-specific model is necessary for the later population.

Appendix: Figures and Tables

Tables

Table 1: Sample characteristics (datA, n = 2000). Continuous variables reported as Mean (SD); categorical variables as Level: n (%).

Variable	Summary
Continuous	
Ki67	21.85 (11.58)
age	54.99 (11.65)
bmi	26.06 (4.56)
log_PFS	2.2 (0.78)
lymph_nodes	2.62 (2.59)
tumor_size	3.38 (1.71)
Categorical	
ER	0: 719 (36%); 1: 1281 (64%)
HER2	0: 1556 (77.8%); 1: 444 (22.2%)
PR	0: 1039 (52%); 1: 961 (48%)
chemo	0: 895 (44.8%); 1: 1105 (55.2%)
hormonal_therapy	0: 1066 (53.3%); 1: 934 (46.7%)
menopause	0: 689 (34.5%); 1: 1311 (65.6%)
race	1: 1284 (64.2%); 2: 289 (14.4%); 3: 176 (8.8%); 4: 251 (12.6%)
stage	1: 504 (25.2%); 2: 893 (44.6%); 3: 603 (30.1%)
surgery	0: 986 (49.3%); 1: 1014 (50.7%)
tumor_grade	1: 383 (19.1%); 2: 871 (43.5%); 3: 746 (37.3%)

Table 2: Cross-validated performance of candidate models (datA).

Model	CV_RMSE	CV_R2
ElasticNet	0.7013	0.2013
PLS	0.7026	0.1974
MARS	0.7034	0.1967
GAM	0.6985	0.2073

Table 3: Non-zero Lasso coefficients (datA).

Variable	Coefficient
(Intercept)	2.5821
ER1	0.2254
HER21	-0.1459
hormonal_therapy1	0.1209
tumor_grade3	-0.1045
lymph_nodes	-0.0599
tumor_size	-0.0481
race4	-0.0167
race3	0.0108
Ki67	-0.0041
age	-0.0019

Table 4: MARS basis function coefficients (datA).

Term	Coefficient
(Intercept)	2.0441
h(6-lymph_nodes)	0.0786
ER1	0.2605
h(bmi-23.9)	-0.0209
h(23.9-bmi)	-0.0500
HER21	-0.1857
hormonal_therapy1	0.1370
tumor_grade3	-0.1704
h(tumor_size-0.4)	-0.0488

Table 5: Generalization performance of Model A on datB.

Model	CV_RMSE_A	RMSE_on_B	R2_on_B	Pct_Increase
ElasticNet	0.7013	0.8361	0.1546	19.2%
PLS	0.7026	0.8361	0.1491	19%
MARS	0.7034	0.8419	0.1573	19.7%
GAM	0.6985	0.8432	0.1620	20.7%

Table 6: Mean (SD) comparison of continuous variables between cohorts.

Variable	Cohort A (1990-1995)	Cohort B (2000-2005)
age	54.99 (11.65)	56.05 (11.22)
bmi	26.06 (4.56)	27.44 (5.05)
tumor_size	3.38 (1.71)	3.08 (1.68)
lymph_nodes	2.62 (2.59)	2.32 (2.53)
Ki67	21.85 (11.58)	20.37 (11.71)
log_PFS	2.2 (0.78)	2.74 (0.74)

Table 7: Cross-validated performance of models on datB.

Model	CV_RMSE	CV_R2
ElasticNet	0.6713	0.1775
PLS	0.6711	0.1773
MARS	0.6717	0.1761
GAM	0.6686	0.1833

Table 8: Elastic net coefficient comparison between cohorts.

Variable	Cohort_A	Cohort_B	Change
(Intercept)	2.5821	3.1115	0.5294
HER21	-0.1459	0.0268	0.1727
race4	-0.0167	0.0734	0.0901
ER1	0.2254	0.1376	-0.0878
stage3	0.0000	-0.0786	-0.0786
PR1	0.0000	0.0711	0.0711
tumor_grade2	0.0000	0.0691	0.0691
hormonal_therapy1	0.1209	0.1685	0.0476
race3	0.0108	-0.0278	-0.0386
lymph_nodes	-0.0599	-0.0229	0.0370
menopause1	0.0000	-0.0328	-0.0328
race2	0.0000	-0.0296	-0.0296
surgery1	0.0000	-0.0206	-0.0206
chemo1	0.0000	-0.0145	-0.0145
tumor_grade3	-0.1045	-0.0954	0.0091
tumor_size	-0.0481	-0.0407	0.0074
bmi	0.0000	-0.0042	-0.0042
stage2	0.0000	-0.0023	-0.0023
age	-0.0019	-0.0026	-0.0007
Ki67	-0.0041	-0.0048	-0.0007

Figures

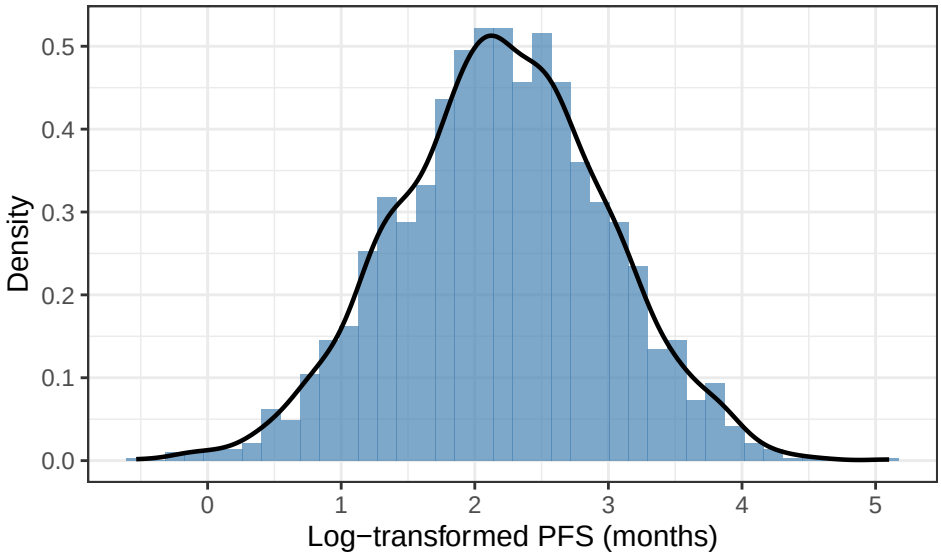

Figure 1: Distribution of log-transformed PFS (datA).

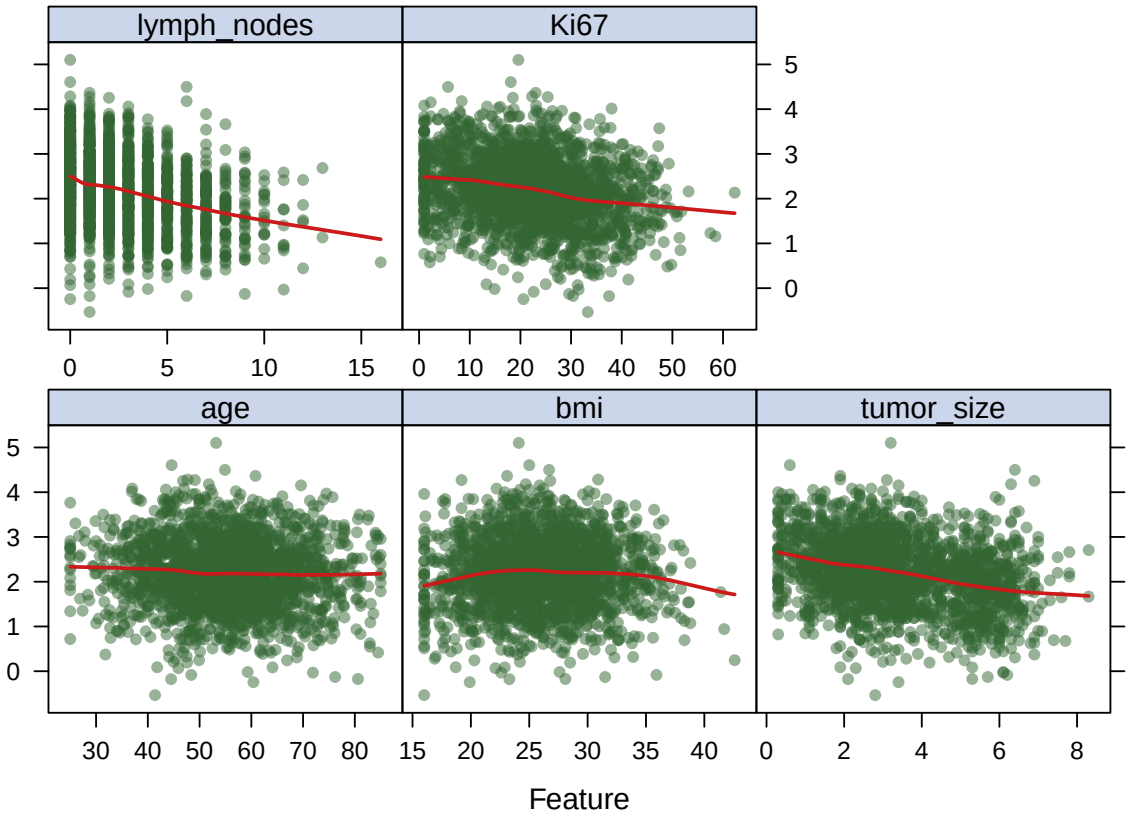

Figure 2: Scatter plots of continuous predictors vs. log_PFS.

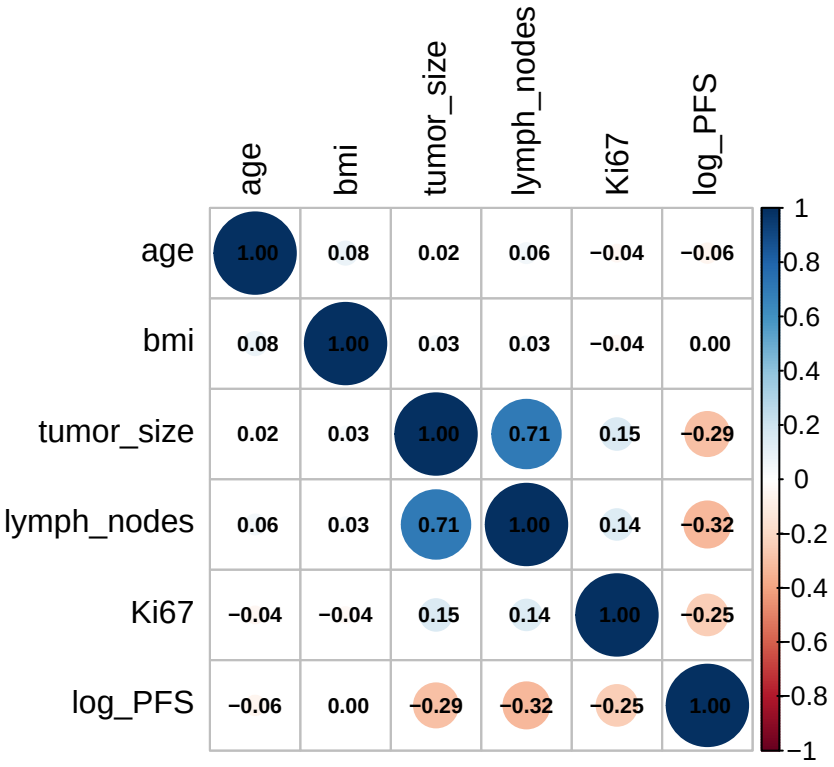

Figure 3: Correlation matrix of continuous variables.

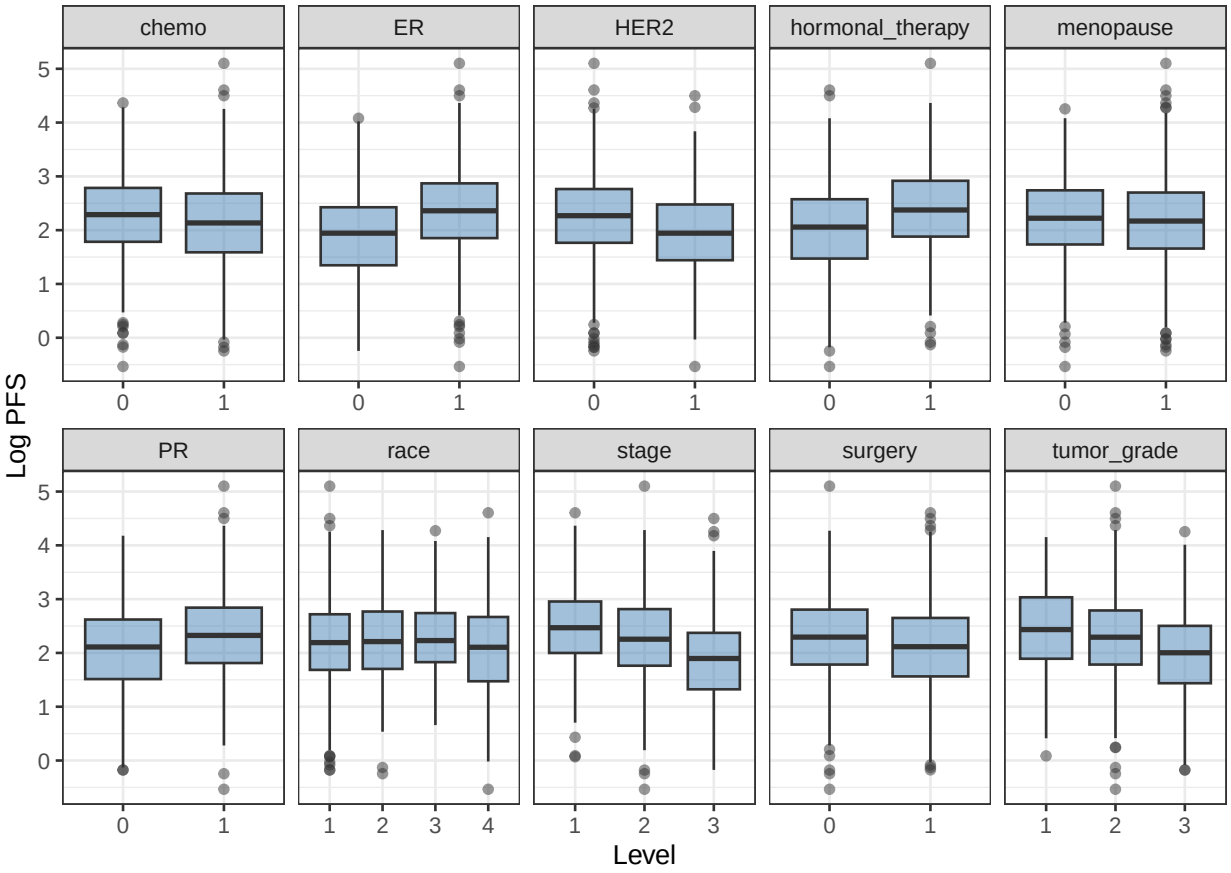

Figure 4: Boxplots of log_PFS by categorical predictors.

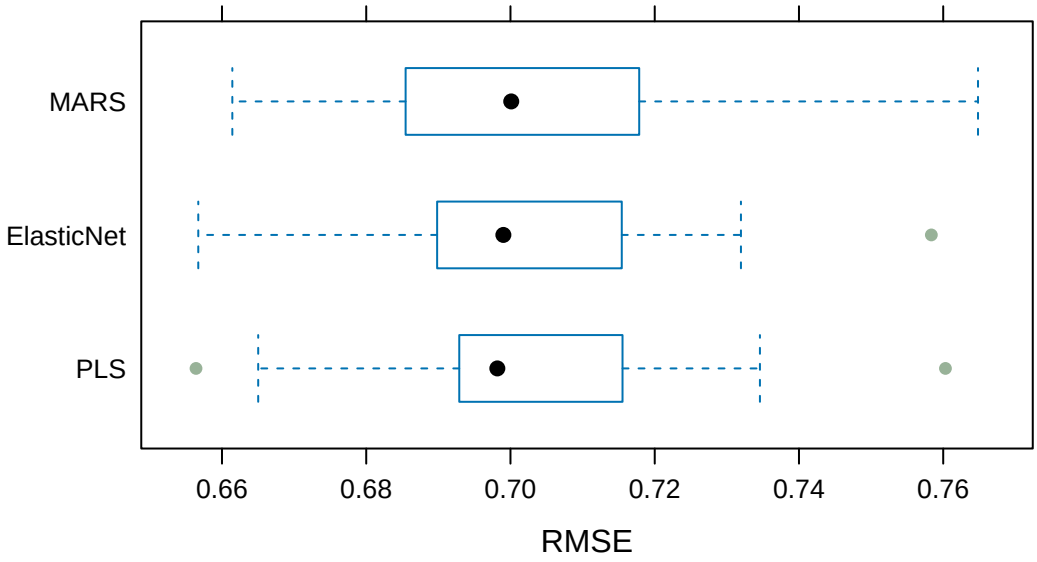

Figure 5: Cross-validated RMSE comparison (datA).

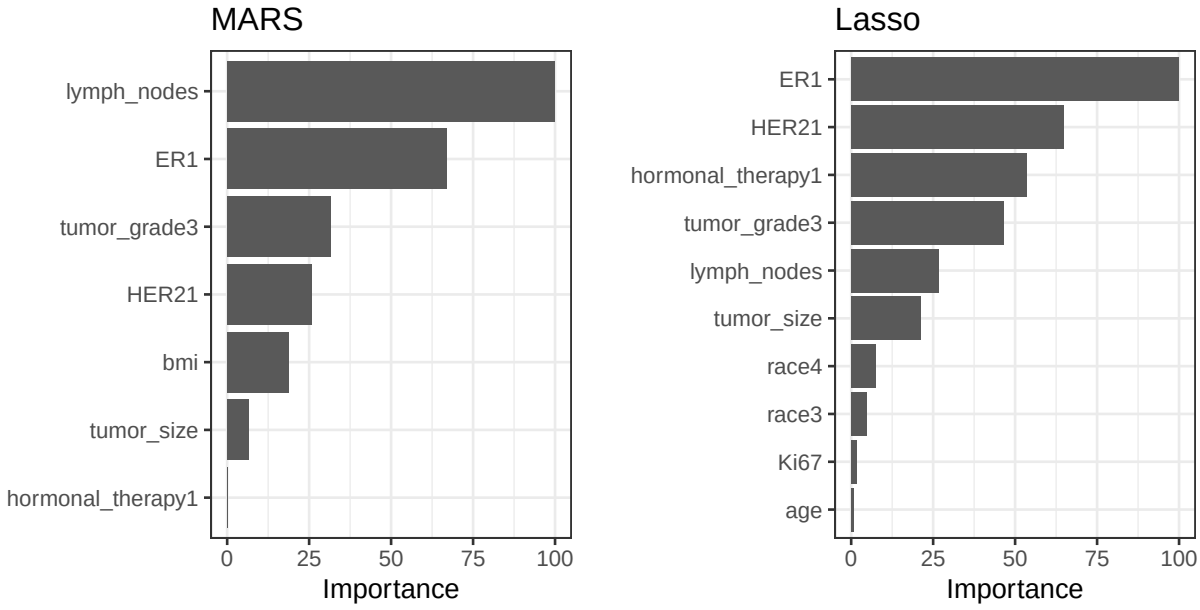

Figure 6: Variable importance from MARS and Lasso (datA).

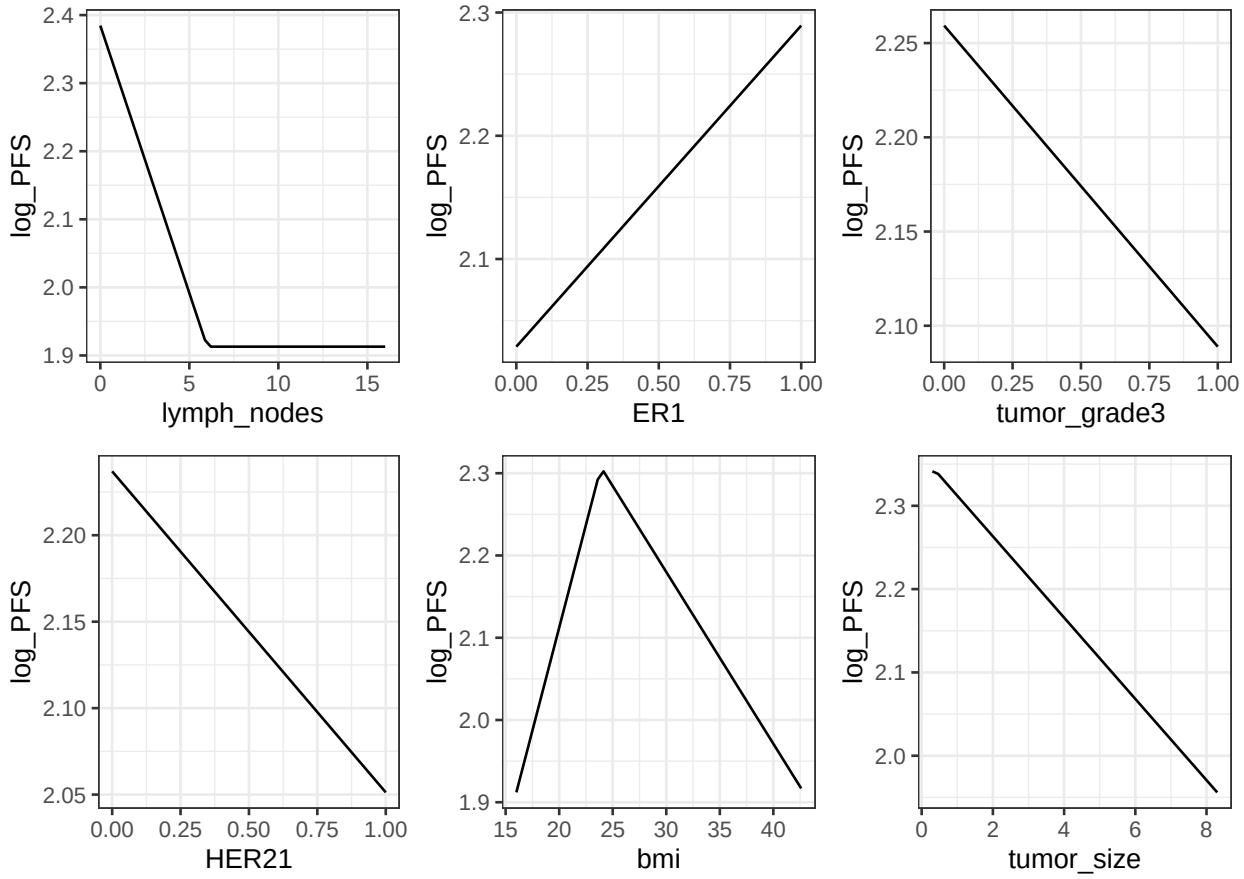

Figure 7: Partial dependence plots for top 6 predictors (MARS, datA).

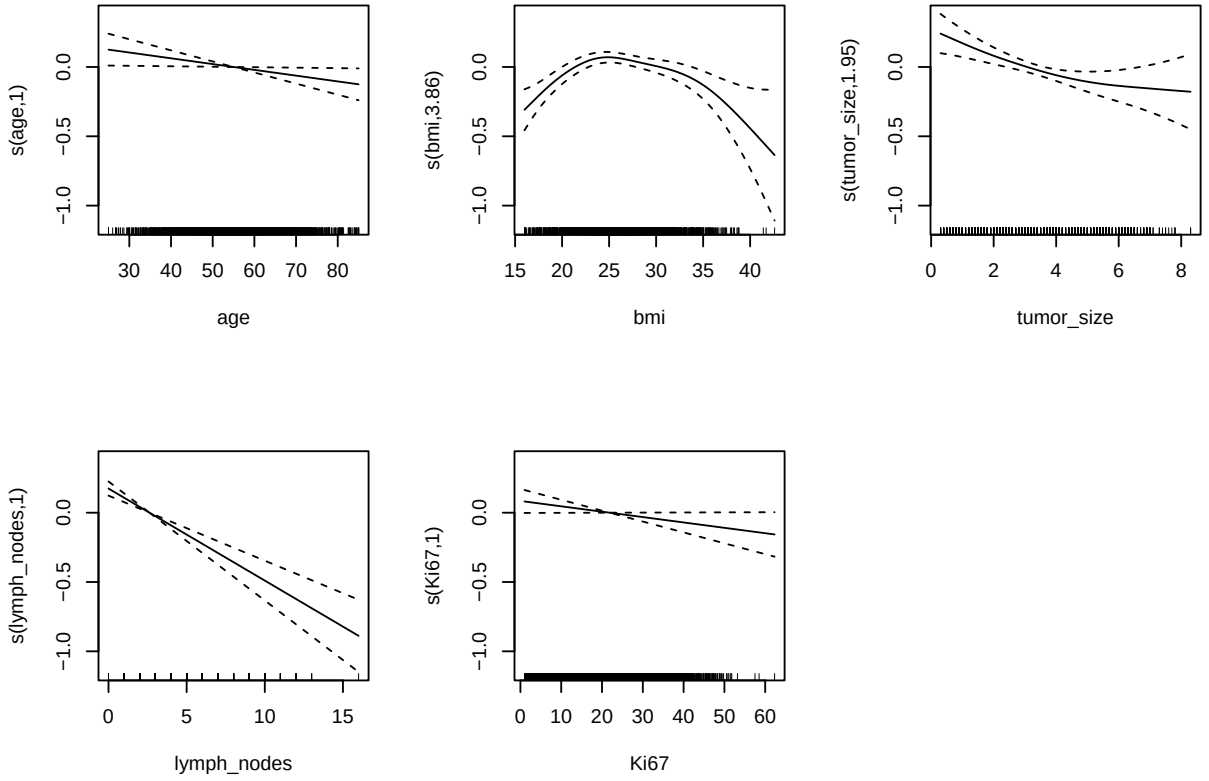

Figure 8: GAM smooth function estimates (datA).

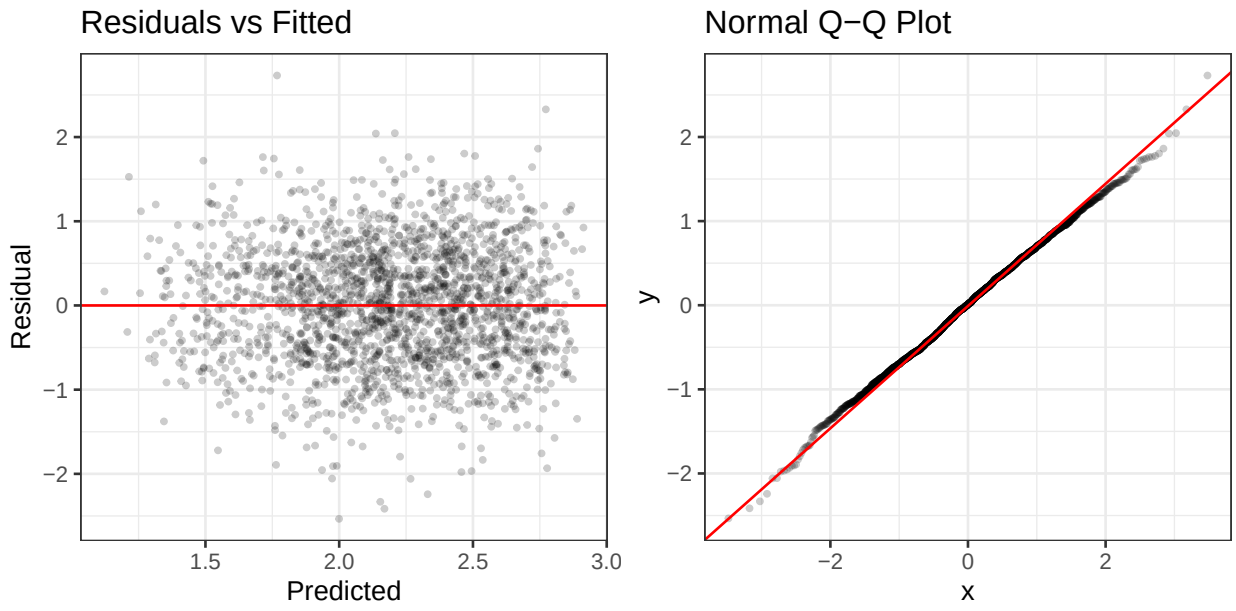

Figure 9: Residual diagnostics for MARS on training data.

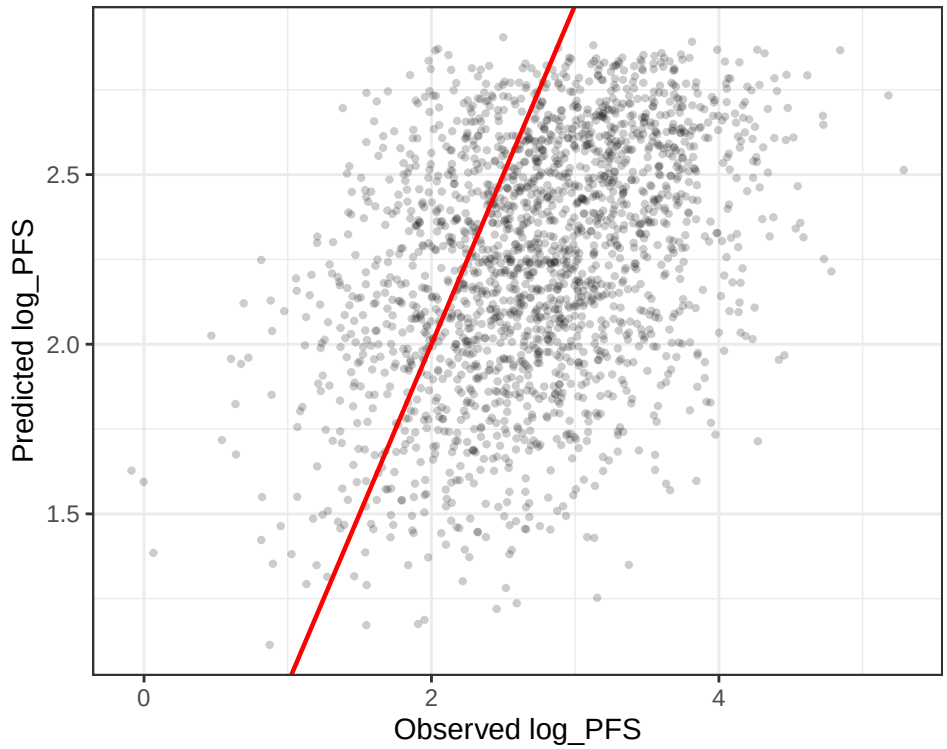

Figure 10: Observed vs. predicted log_PFS on datB (MARS from cohort A).

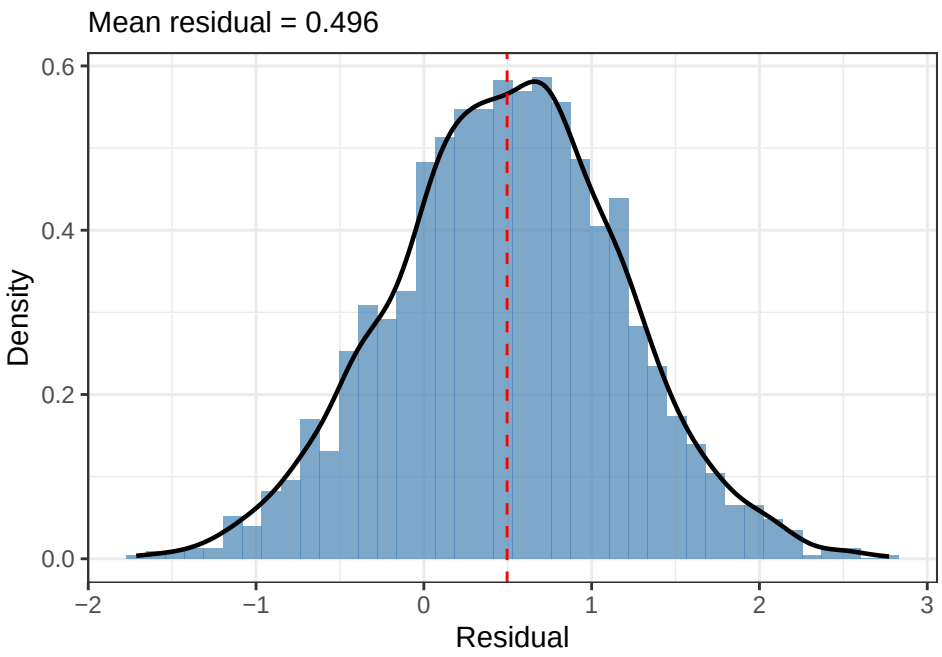

Figure 11: Residual distribution of Model A applied to datB.

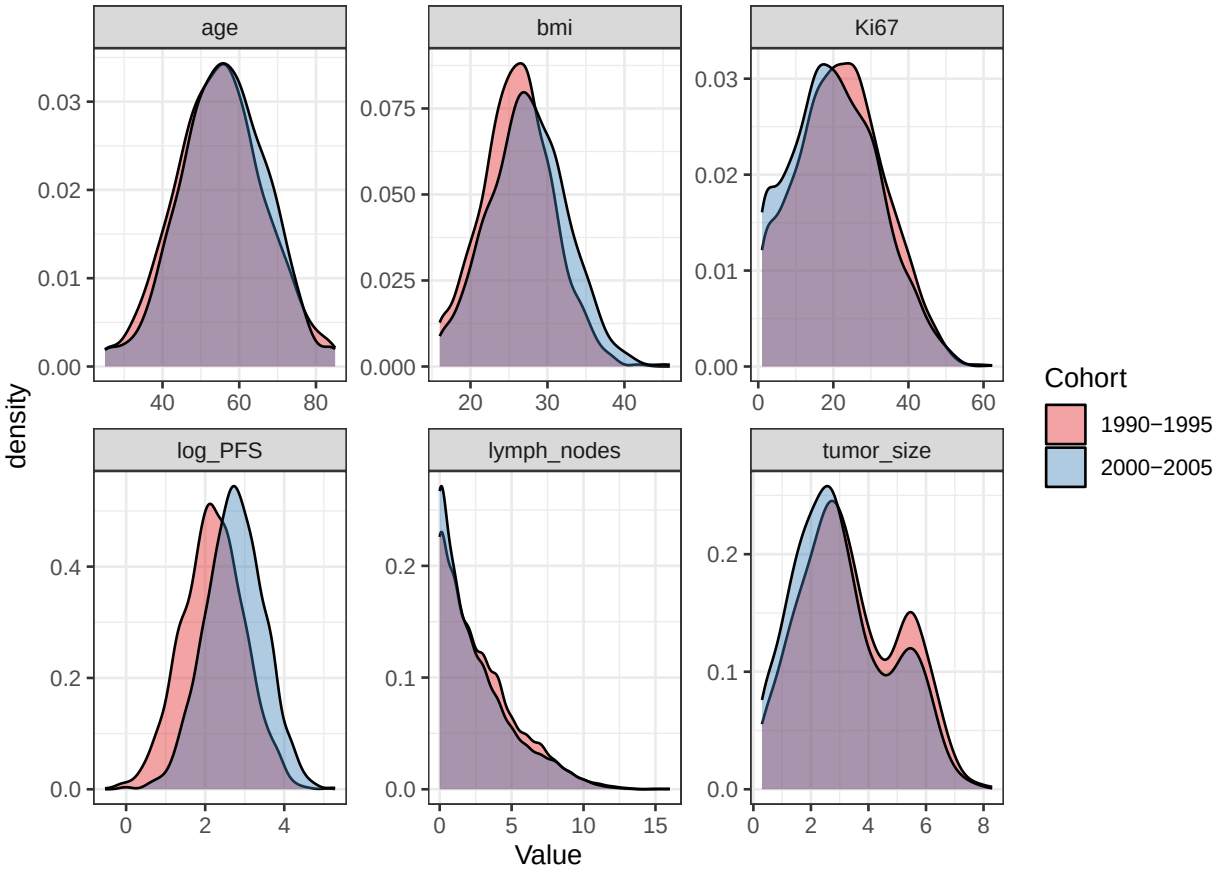

Figure 12: Distribution comparison of continuous variables between cohorts.

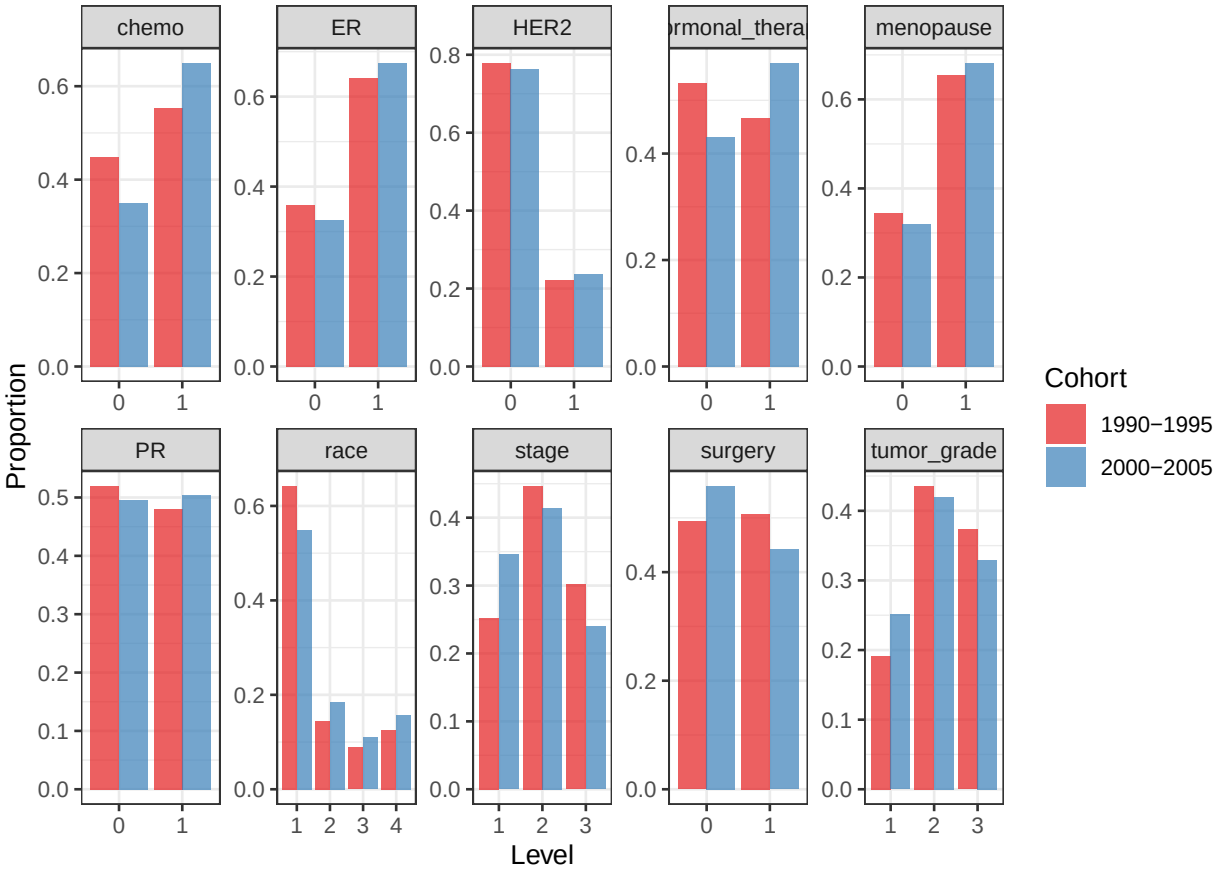

Figure 13: Categorical variable distributions across cohorts.

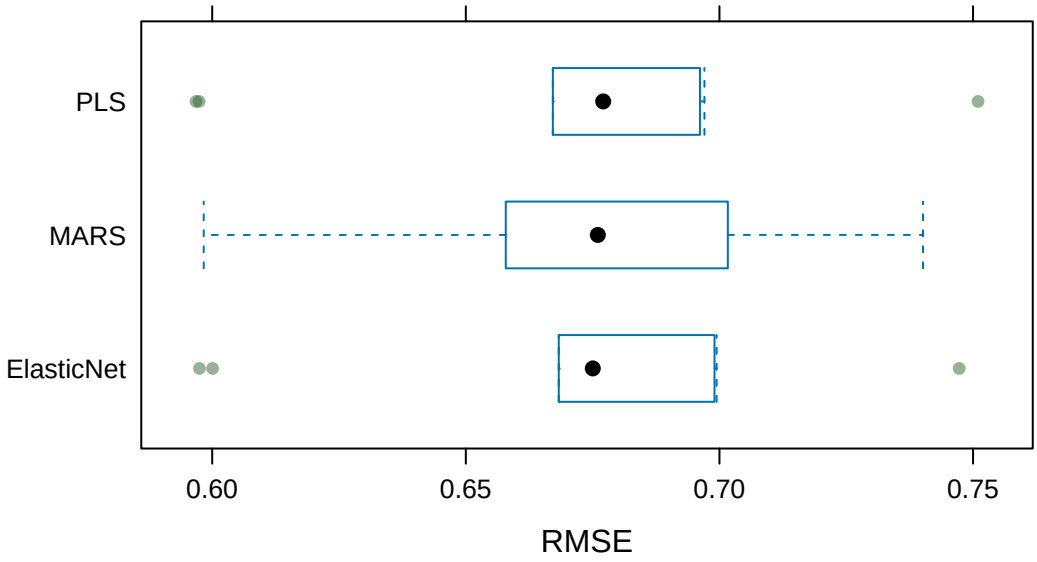

Figure 14: CV RMSE comparison for models trained on datB.

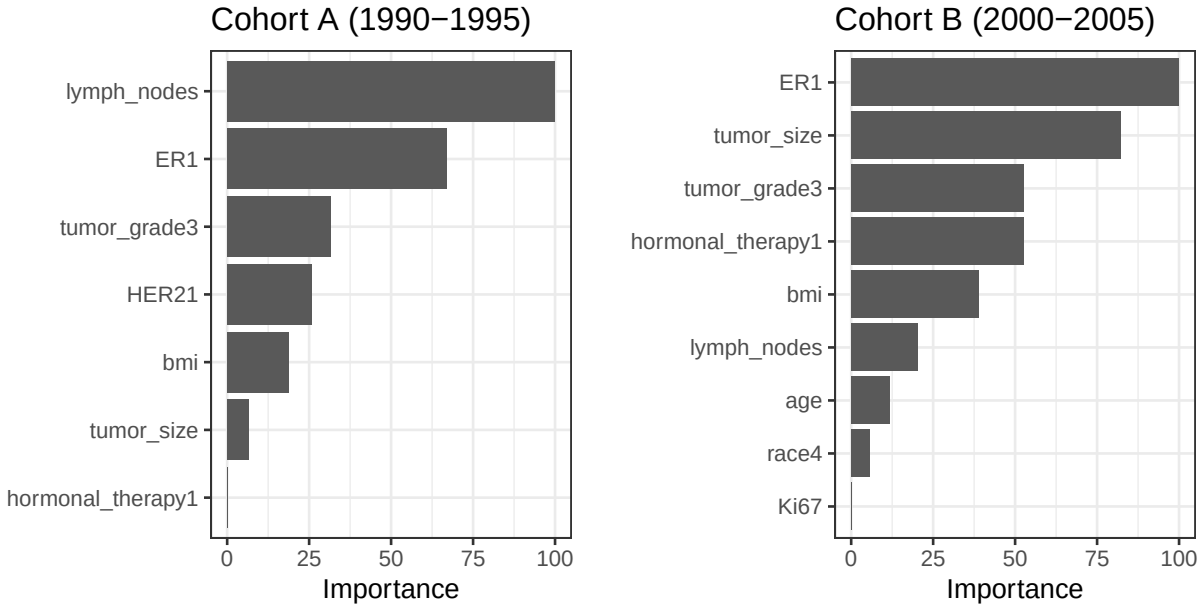

Figure 15: Variable importance comparison (MARS) between cohorts.